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3c21

Ans1)

Foodorder.java

```
package food.model;
```

```
public abstract class FoodOrder {
```

```
    private int orderId;
```

```
    private String customerName;
```

```
    private double amount;
```

```
    static String restaurantName = "Food Express";
```

```
    static int orderCounter = 0;
```

```
    public FoodOrder(int orderId, String customerName, double amount) {
```

```
        this.orderId = orderId;
```

```
        this.customerName = customerName;
```

```
        this.amount = amount;
```

```
        orderCounter++;
```

```
    }
```

```
    public int getOrderId() {
```

```

        return orderId;
    }

    public String getCustomerName() {
        return customerName;
    }

    public double getAmount() {
        return amount;
    }

    public void setOrderId(int orderId) {
        this.orderId = orderId;
    }

    public void setCustomerName(String customerName) {
        this.customerName = customerName;
    }

    public void setAmount(double amount) {
        this.amount = amount;
    }

    public abstract double calculateDeliveryCharge();

    public static void displayTotalOrders() {
        System.out.println("Total Orders: " + orderCounter);
    }
}

```

Discountable.java

```

package food.model;

public interface Discountable {
    double applyDiscount();
}

```

RegularOrder.java

```
package food.model;

public class RegularOrder extends FoodOrder implements Discountable {

    public RegularOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    @Override
    public double calculateDeliveryCharge() {
        return 80;
    }

    @Override
    public double applyDiscount() {
        return getAmount() * 0.10;
    }
}
```

PremiumOrder.java

```
package food.model;

public class PremiumOrder extends FoodOrder implements Discountable {

    public PremiumOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    @Override
    public double calculateDeliveryCharge() {
        return 50;
    }

    @Override
    public double applyDiscount() {
        return getAmount() * 0.15;
    }
}
```

```
}  
}
```

OrderUtility.java

```
package food.utility;  
  
import food.model.FoodOrder;  
import food.model.Discountable;  
  
public class OrderUtility {  
  
    public static boolean validateAmount(double amount) {  
        return amount > 0;  
    }  
  
    public static boolean validateCustomerName(String name) {  
        return name != null && !name.isEmpty();  
    }  
  
    public static void generateOrderSummary(FoodOrder order) {  
        double discount = ((Discountable) order).applyDiscount();  
        double deliveryCharge = order.calculateDeliveryCharge();  
        double finalAmount =  
            order.getAmount() - discount + deliveryCharge;  
  
        System.out.println("Order ID: " + order.getId());  
        System.out.println("Customer: " + order.getCustomerName());  
        System.out.println("Bill Amount: Rs. " + order.getAmount());  
        System.out.println("Discount: Rs. " + discount);  
        System.out.println("Delivery Charge: Rs. " + deliveryCharge);  
        System.out.println("Final Payable: Rs. " + finalAmount);  
    }  
}
```

OrderService.java

```

package food.service;

import food.model.FoodOrder;
import food.utility.OrderUtility;

public class OrderService {

    public void processOrders(FoodOrder[] orders) {

        for (FoodOrder order : orders) {

            if (OrderUtility.validateAmount(order.getAmount())

                && OrderUtility.validateCustomerName(order.getCustomerName())) {

                OrderUtility.generateOrderSummary(order);

            } else {

                System.out.println("Invalid Order: " + order.getId());

            }

        }

    }

}

```

Main.java

```

package food.main;

import food.model.FoodOrder;
import food.model.RegularOrder;
import food.model.PremiumOrder;
import food.service.OrderService;

public class Main {

    public static void main(String[] args) {

        FoodOrder[] orders = new FoodOrder[6];

        orders[0] = new RegularOrder(101, "Rahul", 500);

        orders[1] = new PremiumOrder(102, "Aman", 1000);

    }

}

```

```

orders[2] = new RegularOrder(103, "Priya", 800);
orders[3] = new PremiumOrder(104, "Neha", 1200);
orders[4] = new RegularOrder(105, "Rohan", 600);
orders[5] = new PremiumOrder(106, "Simran", 1500);

OrderService service = new OrderService();
service.processOrders(orders)
FoodOrder.displayTotalOrders();
}
}

```

Ans2)

LibraryResource.java

```

package library.model;

public abstract class LibraryResource {

    private int resourceId;
    private String title;
    private String author;
    private static String libraryName = "University Smart Library";
    private static int resourceCounter = 0;

    public LibraryResource(int resourceId, String title, String author) {
        this.resourceId = resourceId;
        this.title = title;
        this.author = author;
        resourceCounter++;
    }

    public int getResourceId() {

```

```
        return resourceId;
    }
    public String getTitle() {
        return title;
    }
    public String getAuthor() {
        return author;
    }
    public void setResourceId(int resourceId) {
        this.resourceId = resourceId;
    }
    public void setTitle(String title) {
        this.title = title;
    }
    public void setAuthor(String author) {
        this.author = author;
    }
    public static void displayTotalResources() {
        System.out.println("Total Resources Created: " + resourceCounter);
    }
    public static String getLibraryName() {
        return libraryName;
    }
    public abstract double calculateFine(int overdueDays);
    protected void displayBasicDetails() {
        System.out.println("Library: " + libraryName);
    }
}
```

```
        System.out.println("Resource ID: " + resourceId);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
    }
}
```

Printable.java

```
package library.model;

public interface Printable {
    void printDetails();
}
```

Book.java

```
package library.model;

public class Book extends LibraryResource implements Printable {
    public Book(int resourceId, String title, String author) {
        super(resourceId, title, author);
    }

    @Override
    public double calculateFine(int overdueDays) {
        return overdueDays * 5;
    }

    @Override
    public void printDetails() {
        displayBasicDetails();
        System.out.println("Resource Type: Book");
        System.out.println("Fine Rate: Rs. 5/day");
    }
}
```

```
}
```

DigitalResource.java

```
package library.model;
```

```
public class DigitalResource extends LibraryResource implements Printable {
```

```
    public DigitalResource(int resourceId, String title, String author) {
```

```
        super(resourceId, title, author);
```

```
    }
```

```
    @Override
```

```
    public double calculateFine(int overdueDays) {
```

```
        return overdueDays * 2;
```

```
    }
```

```
    @Override
```

```
    public void printDetails() {
```

```
        displayBasicDetails();
```

```
        System.out.println("Resource Type: Digital Resource");
```

```
        System.out.println("Fine Rate: Rs. 2/day");
```

```
    }
```

```
}
```

InputValidator.java

```
package library.util;
```

```
public class InputValidator {
```

```
    public static boolean validateResourceId(int resourceId) {
```

```
        return resourceId > 0;
```

```
    }
```

```
    public static boolean validateFineDays(int days) {
```

```
        return days >= 0;
```

```
    }  
}
```

LibraryService.java

```
package library.service;  
  
import library.model.LibraryResource;  
  
public class LibraryService {  
  
    public double calculateTotalFine(LibraryResource[] resources, int[]  
overdueDays) {  
  
        double totalFine = 0;  
  
        for (int i = 0; i < resources.length; i++) {  
  
            totalFine += resources[i].calculateFine(overdueDays[i]);  
  
        }  
  
        return totalFine;  
  
    }  
}
```

Main.java

```
package library.main;  
  
import library.model.LibraryResource;  
import library.model.Book;  
import library.model.DigitalResource;  
import library.service.LibraryService;  
import library.util.InputValidator;  
  
  
public class Main {  
  
    public static void main(String[] args) {  
  
        LibraryResource[] resources = new LibraryResource[5];  
  
        resources[0] = new Book(101, "Java Programming", "James Gosling");  
  
    }  
}
```

```

resources[1] = new DigitalResource(102, "Machine Learning Course",
"Andrew Ng");

resources[2] = new Book(103, "Data Structures", "Mark Allen");

resources[3] = new DigitalResource(104, "Artificial Intelligence", "Stuart
Russell");

resources[4] = new Book(105, "Operating Systems", "Abraham
Silberschatz");

int[] overdueDays = {5, 3, 10, 4, 2};

double totalFine = 0;

for (int i = 0; i < resources.length; i++) {
    if (InputValidator.validateResourceId(resources[i].getResourceId())
        && InputValidator.validateFineDays(overdueDays[i])) {
        resources[i].printDetails();
        double fine = resources[i].calculateFine(overdueDays[i]);
        System.out.println("Overdue Days: " + overdueDays[i]);
        System.out.println("Fine: Rs. " + fine);
        totalFine += fine;
    }
}

System.out.println("Total Fine of All Resources: Rs. " + totalFine);

LibraryResource.displayTotalResources();
}
}

```